

POWERFACTORY

PowerFactory 2024

Technical Reference

PV Panel

ElmPvpanel, TypPvpanel

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The Solar Panel element *ElmPvpanel* is a two-pole DC model which is designed to reproduce the steady state and dynamic behaviour of a PV module. Based on type parameters specified in the *TypPvpanel* an equivalent circuit model is derived that reproduces the non linear current-voltage characteristic of the solar cells. Based on a solar calculation the actual irradiation on the PV panel surface and cell temperature are obtained, which are used to adapt the model parameters to the actual operating conditions.

1.1 Solar Panel Structure

Solar panels are typically composed of multiple solar cells in series as well as integrated bypass diodes which are connected in parallel to groups of solar cells as can be seen in figure 1.1.

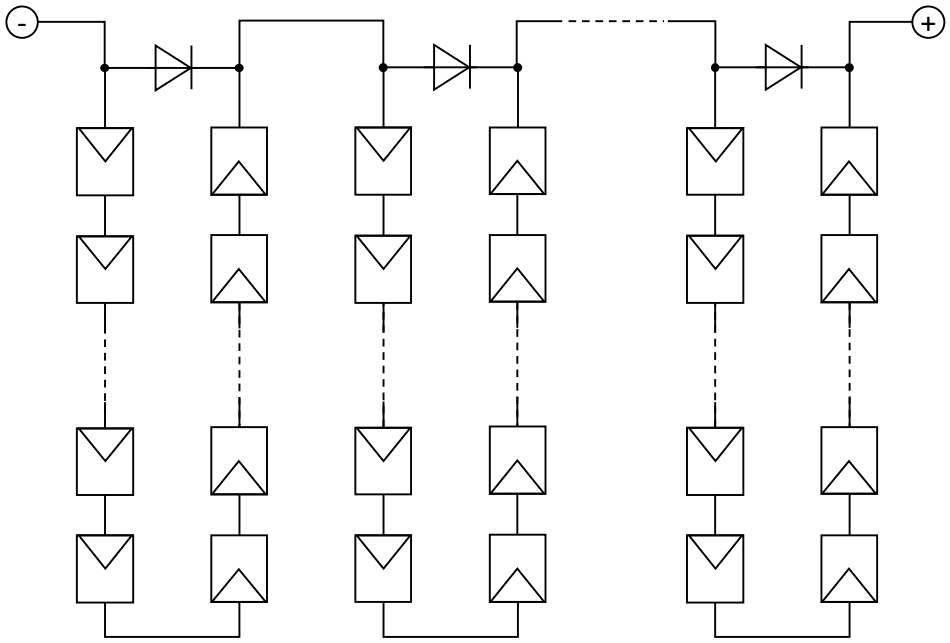


Figure 1.1: Solar panel composed of multiple solar cells in series with integrated bypass diodes

1.2 Steady State Behaviour

The steady state behaviour of a solar panel at given irradiation and temperature is described by a non linear current-voltage characteristic $I(U)$ as shown in figure 1.2. For positive voltages the characteristic is defined by the solar cell behaviour. For negative voltages the characteristic is influenced by the bypass diodes as shown in the figure. The point where the output power reaches its maximum called Maximum Power Point (MPP) and is marked in the curve.

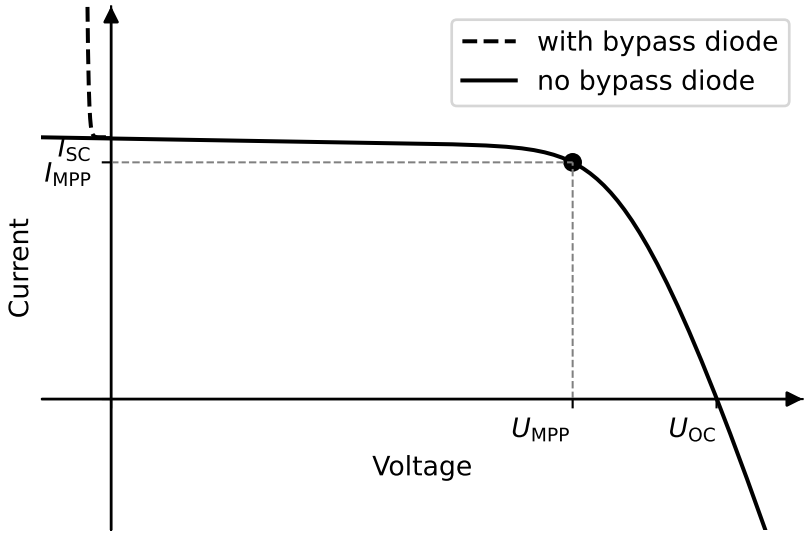


Figure 1.2: Non linear current-voltage characteristic of a solar cell

1.3 Influence of Irradiation and Cell Temperature

The current-voltage characteristic depends on both the actual irradiation $E_{\mathrm{g,pv}}$ on the PV panel surface and the cell temperature T_{C} of the semiconducting material. Both influences are shown in figure 1.3.

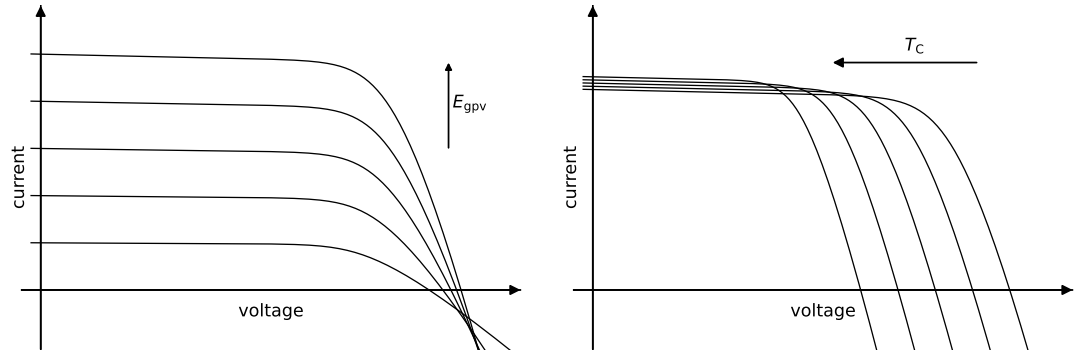


Figure 1.3: Influence of $E_{\mathrm{g,pv}}$ and T_{C} on current-voltage characteristic

With increasing Irradiation the short circuit current of the panel increases, while the open circuit voltage only slightly increases.

With increasing cell temperature the open circuit voltage is reduced significantly (U_{OC}-temperature coefficient $c_{\mathrm{TU}} < 0$) while the short circuit current slightly increases (I_{SC}-temperature coefficient $c_{\mathrm{TI}} > 0$).

1.4 Basic Data

The parameters of the PV panel type (*TypPvpanel*) are shown in Table 1.1.

Table 1.1: Input parameters in Type (*TypPvpanel*)

Parameter	Unit	Default	Description	Range	Symbol
Ppk	W	500	Rated Power of the Panel		$P_{\mathrm{pk,panel}}$
Umpp	V	80	Rated Voltage at MPP		$U_{\mathrm{MPP,panel}}$
Impp	A	6	Rated Current at MPP		$I_{\mathrm{MPP,panel}}$
Uoc	V	90	Open Circuit Voltage		$U_{\mathrm{OC,panel}}$
Isc	A	7	Short-Circuit Current		$I_{\mathrm{SC,panel}}$
material		0	Material		
iusetval		1	Use default values		
cT	%/°C	-0.4	Temperature coefficient (P)	$-100 \ldots 0$	c_{TP}
cTUoc	%/°C	-0.4	Temperature coefficient (U_{OC})	$-100 \ldots 0$	c_{TU}
cTIsc	%/°C	-0.4	Temperature coefficient (I_{SC})	$0 \ldots 100$	c_{TI}
noct	°C	45	Nominal Operating Cell Temperature		T_{NOCT}

The parameters on the Basic Data page of the PV panel element (*ElmPvpanel*) are shown in Table 1.2.

Table 1.2: Input parameters in Basic Data Page (*ElmPvpanel*)

Parameter	Unit	Default Value	Description	Symbol
General Tab				
typ_id			Type	
mode_pgi		2 (Radiation on PV panel)	Model	
npnum		1	Parallel panels	N_{P}
nsnum		1	Panels in series	N_{S}
System Configuration Tab (only needed if mode_pgi == 1)				
cGPSLat	deg		Latitude	ϕ
cGPSLon	deg		Longitude	λ
timezone		UTC+01:00	Local Time Zone	λ_{R}
mount		0 (Fixed/stationary)	Mounting System	
orient	deg	0	Orientation Angle	α
tilt	deg	30	Tilt Angle	β

The orientation angle in the dialogue translates to cardinal directions as follows:

- In the northern hemisphere: 0° ~south, 90° ~west, -90° ~east
- In the southern hemisphere: 0° ~north, 90° ~west, -90° ~east

2 Calculation / Specification of Irradiation on Panel Surface

For calculation of the $I(U)$ characteristic and the electrical model parameters, the global irradiation $E_{\mathrm{g,pv}}$ on the PV panel surface is required.

The user can specify the irradiation by choosing one of two different options for the *Model*-parameter on the Basic Data page:

1. *Solar calculation* (indirect specification of $E_{\mathrm{g,pv}}$)
2. *Irradiation on PV panel* (direct specification of $E_{\mathrm{g,pv}}$)

In the former case the $E_{\mathrm{g,pv}}$ is calculated based on irradiation on the horizontal plane and the panels location, orientation and tilt assuming a perfectly spherical globe. In the latter case $E_{\mathrm{g,pv}}$ is specified directly. The different irradiation components are listed in Table 2.1.

Table 2.1: Irradiation components

Parameter	Act. Value	Description	Comment
-	Egh_a[1]	Global, on horizontal plane	only for solar calculation
-	Ebh_a[1]	Direct (beam), on horizontal plane	only for solar calculation
-	Edh_a[1]	Diffuse, on horizontal plane	only for solar calculation
Egpv[2]	**Egpv_a**	Global, on PV panel surface	
-	Ebpv_a	Direct (beam), on PV panel surface	only for solar calculation
-	Edpv_a	Diffuse, on PV panel surface	only for solar calculation
-	Erpv_a	Reflected, on PV panel surface	only for solar calculation

[1] If *Solar calculation* is selected, two out of three components need to be specified
[2] Input parameter / input signal signal if *Irradiation on PV panel* is selected as input

2.1 Specify indirectly choosing 'Solar Calculation'

On the Load Flow page the user can specify different irradiation components depending on the available data, e.g. global horizontal and global direct irradiation. $E_{\mathrm{g,pv}}$ is calculated based on the panel location, orientation angle and tilt angle, which are specified in the System Configuration subpage of the Basic Data page.

A detailed description of the solar calculation is found in the Technical Reference of the PV System.

Note: If irradiation data from measurement or external sources is used, special attention is recommended when defining the parameters on the System Configuration subpage of the Basic Data page. The irradiation data needs to be consistent with selected coordinates and time zone. Significant deviations might occur if the data is shifted in time.

Note: The solar calculator assumes a perfectly spherical globe without taking into account terrain! If irradiation data measured at an elevated position is used as input, unrealistically high values of $E_{\mathrm{g,pv}}$ might occur in the morning and/or in the evening, especially for high tilt angles.

2.2 Specify directly choosing 'Irradiation on PV panel'

Irradiation on the PV panel surface can be specified directly. This input mode does not require specification of location and time zone and avoids assuming a perfectly spherical globe. It is recommended if:

- the data on the Panel surface is known, e.g. from measurement or external calculations

- the panels are mounted with a high tilt angle,
- terrain and/or shading significantly influence the irradiation on the panel surface or
- time domain simulations (RMS, EMT) with variable irradiation are performed.

3 Load Flow Analysis

3.1 Equivalent Circuit Model

A common approach to model this behaviour is to use a circuit model, that represents the physical effects inside the solar cells. Those models are usually composed of an irradiation dependent current source, one or more diodes to represent the semiconducting cell material as well as resistors to model the current and voltage dependent losses inside the semiconductor and the wiring.

In *PowerFactory* the common single diode five-parameter-model was implemented due to its combination of good accuracy and convenient parameterisation. It is complemented with an optional bypass model including a diode and a small resistor. The resulting equivalent circuit model is shown in figure 3.1.

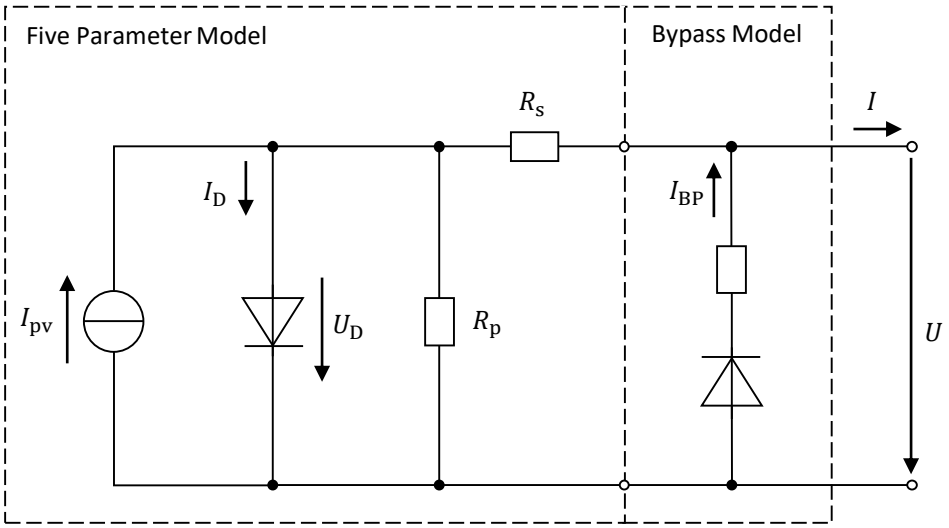


Figure 3.1: Five parameter model of photovoltaic panel with optional bypass model

3.2 Model Equations

The PV panel current I is calculated as

$$\begin{aligned} I &= I_{\mathrm{PV}} - I_{\mathrm{D}} - I_{\mathrm{Rp}} + I_{\mathrm{BP}} \\ &= I_{\mathrm{PV}} - I_0 \cdot \left(\mathrm{e}^{\frac{U_{\mathrm{D}}}{U_{\mathrm{T}}}} - 1 \right) - \frac{U_{\mathrm{D}}}{R_{\mathrm{P}}} + I_{\mathrm{BP}} \end{aligned} \tag{1}$$

with the diode voltage

$$U_{\mathrm{D}} = U + R_{\mathrm{S}} \cdot (I - I_{\mathrm{BP}}). \tag{2}$$

If the bypass is not considered ($useBP = 0$) the bypass current is set to $I_{\mathrm{BP}} = 0$. If the bypass diode is considered, the user can choose between a linearised bypass model (equ. 3)

$$I_{\mathrm{BP}} = \begin{cases} -\frac{U - U_{\mathrm{f,BP}}}{R_{\mathrm{on,BP}}} & \text{if } U < -U_{\mathrm{f,BP}} \text{ (bypass conducts)} \\ -U \cdot G_{\mathrm{off,BP}} & \text{else (bypass diode blocks)} \end{cases} \quad (3)$$

or an exponential diode function (equ. 4)

$$I_{\mathrm{BP}} = I_{0,\mathrm{BP}} \cdot \left(\mathrm{e}^{\frac{-(U + R_{\mathrm{on,BP}} \cdot I_{\mathrm{BP}})}{U_{\mathrm{T,BP}}}} - 1 \right) \quad (4)$$

where the parameters $I_{0,\mathrm{BP}}$ and $U_{\mathrm{T,BP}}$ are calculated based on $U_{\mathrm{f,BP}}$.

The five model parameters are

- Photovoltaic source current I_{PV},
- Diode saturation current I_0,
- Diode ideality factor A in $U_{\mathrm{T}} = \frac{k \cdot A \cdot T \cdot N_{\mathrm{C}}}{q}$ with cell temperature T in K, Boltzmann constant k , number N_{C} of serial cells per panel and elementary charge q,
- Series resistor R_{S} and
- Parallel resistor R_{P}

complemented by the optional bypass parameters

- Forward voltage $U_{\mathrm{F,BP}}$,
- On-resistance $R_{\mathrm{on,BP}}$ and
- Off-conductance $G_{\mathrm{off,BP}}$ of the bypass.

3.3 Parameter Calculation

3.3.1 Given data and preliminary calculations

Manufacturers usually provide points on the I(U) characteristic for standard test conditions (STC, 1000 W/m^2 irradiation on the panel surface and 25 °C cell temperature):

- Open circuit voltage $U_{\mathrm{OC,panel}}$
- Short circuit voltage $I_{\mathrm{SC,panel}}$
- Voltage at maximum power point $U_{\mathrm{MPP,panel}}$
- Current at maximum power point $I_{\mathrm{MPP,panel}}$

These parameters, as well temperature coefficients, cell material and normal operating conditions, are defined in the PV panel type (TypPvpanel).

In addition to these parameters, the number $N_{\mathrm{C,panel}}$ of cells in series per panel is required. Since this information is not always provided in manufacturers data sheets, it is estimated based on the ratio $U_{\mathrm{OC,panel}}/U_{\mathrm{OC,cell}}$. The open circuit voltage per cell depends on the material and is assumed acc. to table 3.1.

Table 3.1: Assumed open circuit voltage ranges of a single solar cell depending on the cell material

Material	$U_{\mathrm{OC,cell}}$ in V
Mono-crystalline silicon	0.55 . . . 0.68
Poly-crystalline silicon	0.50 . . . 0.65
Amorphous silicon	0.70 . . . 1.10
Copper-indium-selenide	0.50 . . . 0.70
Cadmium telluride	0.80 . . . 1.00
Other	0.55 . . . 0.68

The user can define the model to have multiple panels in series and/or in parallel. One overall circuit model according to figure 3.1 is created to represent all N_{S} serial and N_{P} parallel panels. To do so, the panel type data is translated to higher numbers of panels as follows:

$$U_{\mathrm{OC}} = N_{\mathrm{S}} \cdot U_{\mathrm{OC,panel}} \tag{5}$$

$$U_{\mathrm{MPP}} = N_{\mathrm{S}} \cdot U_{\mathrm{MPP,panel}} \tag{6}$$

$$N_{\mathrm{C}} = N_{\mathrm{S}} \cdot N_{\mathrm{C,panel}} \tag{7}$$

$$I_{\mathrm{SC}} = N_{\mathrm{P}} \cdot I_{\mathrm{SC,panel}} \tag{8}$$

$$I_{\mathrm{MPP}} = N_{\mathrm{P}} \cdot I_{\mathrm{MPP,panel}} \tag{9}$$

3.3.2 Parameter Calculation for Standard Test Conditions

The five model parameters are calculated iteratively, using the algorithms acc. to [3].

In the first step R_{P} is neglected. The remaining parameters are estimated as follows, with the starting value of the diode ideality factor $A = 1$:

$$U_{\mathrm{T}} = \frac{k \cdot A \cdot T \cdot N_{\mathrm{C}}}{q} \tag{10}$$

$$R_{\mathrm{S}} = \frac{U_{\mathrm{OC}}}{I_{\mathrm{MPP}}} + \frac{U_{\mathrm{T}}}{I_{\mathrm{MPP}}} \ln\left(\frac{U_{\mathrm{T}}}{U_{\mathrm{T}} U_{\mathrm{MPP}}}\right) - \frac{U_{\mathrm{MPP}}}{I_{\mathrm{MPP}}} \tag{11}$$

$$I_0 = \frac{I_{\mathrm{SC}}}{\mathrm{e}^{\frac{U_{\mathrm{OC}}}{U_{\mathrm{T}}}} - \mathrm{e}^{\frac{R_{\mathrm{S}} I_{\mathrm{SC}}}{U_{\mathrm{T}}}}} \tag{12}$$

$$I_{\mathrm{PV}} = I_0 \left(\mathrm{e}^{\frac{U_{\mathrm{OC}}}{U_{\mathrm{T}}}} - 1\right) \tag{13}$$

The MPP voltage $U_{\mathrm{MPP,c}}$, which would result from these parameters, is calculated as

$$U_{\mathrm{MPP,c}} = \left[U_{\mathrm{T}} \ln\left(\frac{I_{\mathrm{PV}} + I_0 - I_{\mathrm{MPP}}}{I_0}\right)\right] - (R_{\mathrm{S}} I_{\mathrm{MPP}}) \tag{14}$$

and compared to the actual value U_{MPP} from the data sheet. If $U_{\mathrm{MPP,c}} < U_{\mathrm{MPP}}$, then in the next iteration A is increased. If $U_{\mathrm{MPP,c}} > U_{\mathrm{MPP}}$ then A is decreased. The procedure is shown as flow chart in figure 3.2

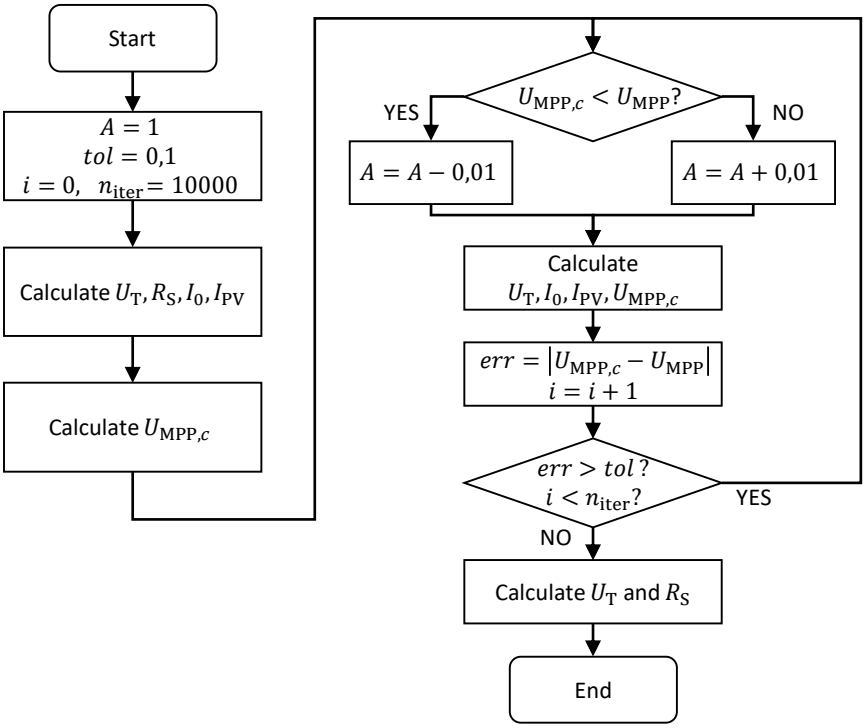


Figure 3.2: First step of parameter determination for A and R_{S} acc. to [3]; calculation of parameters acc. to equations (10) to (14)

In the second step of the parameter determination method R_{P} is calculated in a way to better fit the MPP provided by the manufacturers data sheet which is defined in the PV panel type (TypPvpanel). The maximum output power can be related to the circuit parameters as

$$P_{\max} = U_{\mathrm{MPP}} \cdot I_{\mathrm{MPP}} = U_{\mathrm{MPP}} \cdot \left[I_{\mathrm{PV}} - I_0 \left(\mathrm{e}^{\frac{U_{\mathrm{MPP}} + R_{\mathrm{S}} I_{\mathrm{MPP}}}{U_{\mathrm{T}}}} - 1 \right) - \frac{U_{\mathrm{MPP}} + R_{\mathrm{S}} I_{\mathrm{MPP}}}{R_{\mathrm{P}}} \right]. \quad (15)$$

Based on equation (15) the starting point for the parallel resistance is calculated as

$$R_{\mathrm{P}} = \frac{U_{\mathrm{MPP}}(U_{\mathrm{MPP}} + R_{\mathrm{S}} I_{\mathrm{MPP}})}{U_{\mathrm{MPP}} I_{\mathrm{PV}} - U_{\mathrm{MPP}} I_0 \left(\mathrm{e}^{\frac{U_{\mathrm{MPP}} + R_{\mathrm{S}} I_{\mathrm{MPP}}}{U_{\mathrm{T}}}} - 1 \right) - U_{\mathrm{MPP}} I_{\mathrm{MPP}}}. \quad (16)$$

I_0 and I_{PV} are recalculated, now considering R_{P}:

$$I_0 = \frac{I_{\mathrm{SC}} \left(1 + \frac{R_{\mathrm{S}}}{R_{\mathrm{P}}} \right) - \frac{U_{\mathrm{OC}}}{R_{\mathrm{P}}}}{\mathrm{e}^{\frac{U_{\mathrm{OC}}}{U_{\mathrm{T}}}} - \mathrm{e}^{\frac{R_{\mathrm{S}} I_{\mathrm{SC}}}{U_{\mathrm{T}}}}} \quad (17)$$

$$I_{\mathrm{PV}} = I_0 \left(\mathrm{e}^{\frac{U_{\mathrm{OC}}}{U_{\mathrm{T}}}} - 1 \right) + \frac{U_{\mathrm{OC}}}{R_{\mathrm{P}}} \quad (18)$$

Now $P_{\max}$ is recalculated according to equation (15) and used to calculate the theoretical value $I_{\mathrm{MPP,c}} = P_{\max}/U_{\mathrm{MPP,c}}$ for comparison with the real I_{MPP}. This comparison is then used to iteratively change R_{P}, until a good fit is reached. The entire algorithm is shown in the flow chart in figure 3.3.

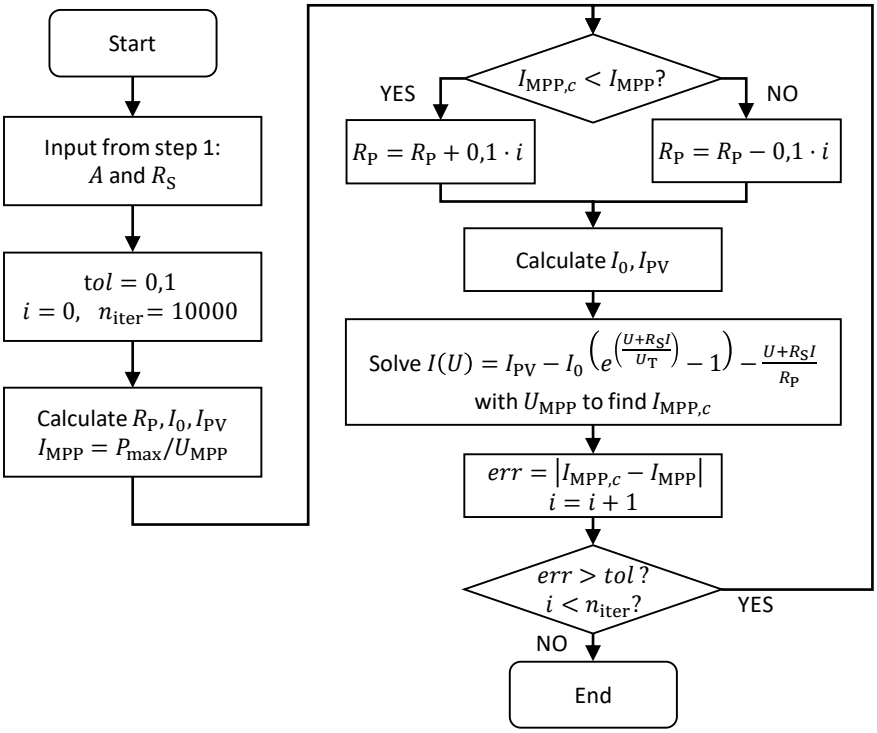


Figure 3.3: Second step of parameter determination for R_{P} acc. to [3]; calculation of parameters acc. to equations (15) to (18)

3.3.3 Translation of Parameters to Other Operating Conditions

The previously described parameters are valid for standard test conditions (STC, 1000 W/m^2 irradiation on the panel surface and 25 °C cell temperature). For other irradiations and cell temperatures the parameters need to be adapted, to achieve a behaviour as described in section 1.3.

In a first step the actual cell temperature is estimated based on the ambient temperature T_{amb}, the normal operating cell temperature $T_{\mathrm{C,NOCT}}$ (at 800 W/m^2 irradiation and 1 W/s wind speed and 20 °C ambient temperature) provided in the data sheet and defined in the type data.

$$T_{\mathrm{C,act}} = T_{\mathrm{amb}} + (T_{\mathrm{C,NOCT}} - 20^{\circ}\mathrm{C}) \cdot \frac{E_{\mathrm{g,pv}}}{800\,\mathrm{W/m^2}} \tag{19}$$

The actual parameters are calculated based on $T_{\mathrm{C,act}}$, the temperature coefficients c_{TI} for short circuit current and c_{TU} for open circuit voltage and the actual relative irradiation $e_{\mathrm{rel}} = \frac{E_{\mathrm{g,pv}}}{E_{\mathrm{STC}}}$

based on [2] with a few adaptations.

$$I_{\mathrm{PV,act}} = e_{\mathrm{rel}} \cdot I_{\mathrm{PV}} \cdot \left[1 + \frac{c_{\mathrm{TI}}}{100\%} \cdot (T_{\mathrm{C,act}} - T_{\mathrm{C,STC}})\right] \tag{20}$$

$$U_{\mathrm{T,act}} = U_{\mathrm{T}} \cdot \frac{T_{\mathrm{C,act}}/\mathrm{K}}{T_{\mathrm{C,STC}}/\mathrm{K}} \tag{21}$$

$$R_{\mathrm{S,act}} = \frac{R_{\mathrm{S}}}{0.9 \cdot e_{\mathrm{rel}} + 0.1} \tag{22}$$

$$R_{\mathrm{P,act}} = \frac{R_{\mathrm{P}}}{e_{\mathrm{rel}}} \tag{23}$$

$$I_{0,\mathrm{act}} = \exp\left[\frac{e_{\mathrm{rel}} - 0.2}{1 - 0.2} \cdot \ln \frac{I_0\,(1,T_{\mathrm{C,act}})}{I_0\,(0.2,T_{\mathrm{C,act}})} + \ln I_0\,(0.2,T_{\mathrm{C,act}})\right] \tag{24}$$

In equation 24 the currents $I_0\,(e_{\mathrm{rel}},T_{\mathrm{C,act}})$ are calculated using equation (17) but with $U_{\mathrm{T,act}}$, $R_{\mathrm{S,act}}$ and $R_{\mathrm{P,act}}$ acc. to equations (21) to (23) and with

$$I_{\mathrm{SC,act}} = e_{\mathrm{rel}} \cdot I_{\mathrm{SC}} \cdot \left[1 + \frac{c_{\mathrm{TI}}}{100\%} \cdot (T_{\mathrm{C,act}} - T_{\mathrm{C,STC}})\right] \quad \text{and} \tag{25}$$

$$U_{\mathrm{OC,act}} = U_{\mathrm{OC}} \cdot \left[1 + \frac{c_{\mathrm{TU}}}{100\%} \cdot (T_{\mathrm{C,act}} - T_{\mathrm{C,STC}})\right] + U_{\mathrm{T,act}} \cdot \ln(e_{\mathrm{rel}}). \tag{26}$$

3.4 Input parameters, signals and state variables

Table 3.2: Input parameters in Load Flow Page (*ElmPvpanel*)

Parameter	Unit	Default	Description	Range	Symbol
Irradiance input and environment factors					
iopt_rad		1	Specified components option		
iopt_glo		0	Global irradiance option		
kt		0.6	Clearness index	$0 \ldots 1$	KT
ghi	W/m^2	600	Global horizontal irradiance		$E_{\mathrm{g,hor}}$
iopt_dir			Direct irradiance option		
dhi	W/m^2	400	Direct horizontal Irradiance		$E_{\mathrm{b,hor}}$
dni	W/m^2	400	Direct normal irradiance		$E_{\mathrm{b,norm}}$
iopt_dif			Diffuse Irradiance option		
Tamb	°C	25	Ambient temperature		T_{a}
shfdir		0	Shadowing factor (Direct)	$0 \ldots 1$	S_{dir}
shfdif		0	Shadowing factor (Diffuse)	$0 \ldots 1$	S_{diff}
albedo		0.31	Ground albedo		ρ_{g}
Egpv	W/m^2	800	Global irradiation on PV panel	$E_{\mathrm{g,pv}} \geq 0$	$E_{\mathrm{g,pv}}$
Bypass diode parameters					
useBpDiode		1	Incl. bypass diode in circuit model		
iopt_bplin		1	Use linearized bypass model		
Uf_BpDiode	V	2.1	Forward voltage		$U_{\mathrm{f,BP}}$
Ron_BpDiode	Ohm	0.1	On resistance		$R_{\mathrm{on,BP}}$
Goff_BpDiode	S	1.e-6	Off conductance		$G_{\mathrm{off,BP}}$

Table 3.3: Input Signals in Load Flow Analysis

Input Signal	Unit	Description	Symbol
Egpv	W/m^2	Global irradiation on PV panel	$E_{\mathrm{g,pv}}$

Table 3.4: State variables in Load Flow Analysis

State	Unit	Type	Description	Symbol
BP_BLOCKED		discrete	Switching state of bypass diode	

4 RMS and EMT Simulation

Changes in the irradiation $E_{\mathrm{g,pv}}$ during EMT/RMS simulation can be applied using parameter events or by connecting a signal to the input. If the irradiation changes, the circuit parameters will be recalculated using the translational equations (20) to (26). The cell temperature will only be calculated once during initialisation based on the initial irradiation. In other words, equation (19) is not updated during EMT/RMS simulation. This is because changes of the irradiation do not influence the temperature immediately, but based on unknown thermal time constants, which are usually significantly higher then typical simulation times.

Note: If the irradiation *Egpv* is changed abruptly during simulation by a connected signal, convergence issues might occur. In this case please consider adapting the irradiation in a continuous way, e.g. using a slope or a by filtering the signal with a first order lag.

4.1 RMS Simulation

The RMS model equations are identical to the Load Flow model.

4.2 EMT Simulation

4.2.1 Equivalent Circuit Model for EMT Simulation

Most model equations, such as the internal and bypass current equations, are identical to the load flow equations. To account for dynamic effects, the EMT model extends the Load Flow and RMS model with a capacitor as shown in figure 4.1. The capacitor current I_{C} is added to the node current equation (1) resulting in

$$I = I_{\mathrm{PV}} - I_{\mathrm{D}} - I_{\mathrm{Rp}} - I_{\mathrm{C}} + I_{\mathrm{BP}}. \tag{27}$$

4.2.2 Constant Capacitor Model

If a constant capacitor is selected, the current is

$$I_{\mathrm{C}} = \frac{\mathrm{d}Q_{\mathrm{Cp}}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(C_{\mathrm{P}} \cdot U_{\mathrm{D}}\right) = C_{\mathrm{P}} \cdot \frac{\mathrm{d}U_{\mathrm{D}}}{\mathrm{d}t}. \tag{28}$$

4.2.3 Voltage and Current Dependent Capacitor Model

Instead of a constant capacitor a voltage and current dependent capacitor acc. to [1] can be selected, which consists of the junction capacitance C_{j} and the diffusion capacitance C_{d}.

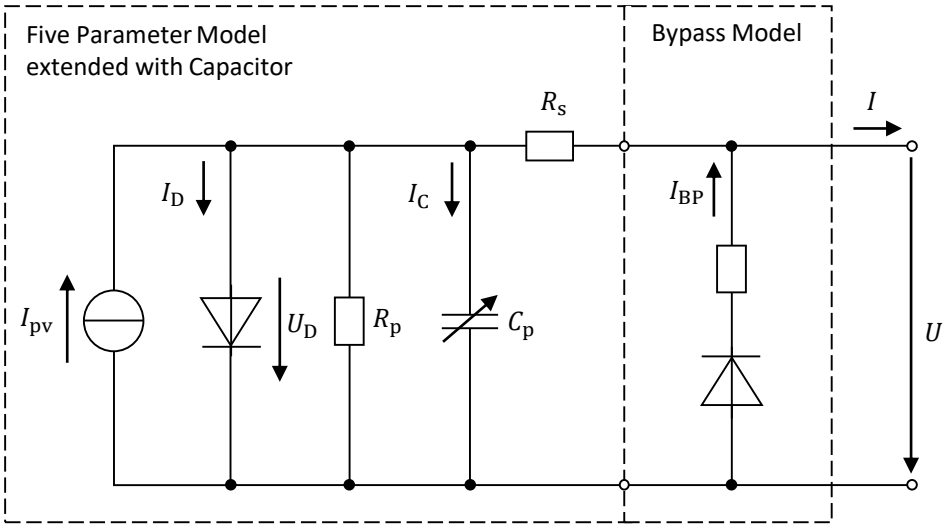


Figure 4.1: Dynamic model of solar cell with optional bypass model, extension of five parameter model

The junction capacitance C_j comes from the charge stored in the depletion region at the semiconductor PN junction. It is voltage dependent and calculated as

$$C_j(U_D) = \frac{C_j0}{N_C\sqrt{1-\frac{U_D}{N_C\phi_0}}}. \tag{29}$$

The zero-voltage capacitance C_{j0} of the PV cell PN junction and the zero-voltage junction potential ϕ_0 are input parameters, while the total number N_C of PV cells in series is estimated based on the panel type data as described in section 3.3.1. The current through the junction capacitor is

$$I_{Cj} = \frac{dQ_{Cj}}{dt} = \frac{d}{dt}(C_j \cdot U_D) = \frac{dC_j}{dt}\cdot U_D + C_j \cdot \frac{dU_D}{dt} \tag{30}$$

with

$$\frac{dC_j}{dt} = \frac{dC_j}{dU_D}\cdot\frac{dU_D}{dt} = \frac{C_{j0}}{\left(2N_C^2\phi_0\cdot(1-U_D/(N_C\cdot\phi_0))^{\frac{3}{2}}\right)}\cdot\frac{dU_D}{dt} \tag{31}$$

The diffusion capacitance C_d comes from the charge stored in the semiconductor neutral region, outside the depletion region, and is current dependent. It is calculated as

$$C_d(I_D) = \frac{\tau\cdot I_D}{U_T} \tag{32}$$

with the mean carrier lifetime τ. Since the diode current is voltage dependent it can be rewritten to

$$C_d(U_D) = \frac{\tau}{U_T}\cdot I_D(U_D) = \frac{\tau}{U_T}\cdot I_0\left(e^{\frac{U_D}{U_T}}-1\right) \tag{33}$$

Its current is

$$I_{Cd} = \frac{dQ_{Cd}}{dt} = \frac{d}{dt}(C_d\cdot U_D) = \frac{dC_d}{dt}\cdot U_D + C_d\cdot\frac{dU_D}{dt} \tag{34}$$

with

$$\begin{aligned}\frac{\mathrm{d}C_\mathrm{d}}{\mathrm{d}t} &= \frac{\mathrm{d}C_\mathrm{d}}{\mathrm{d}U_\mathrm{D}} \cdot \frac{\mathrm{d}U_\mathrm{D}}{\mathrm{d}t} \\ &= \frac{\mathrm{d}}{\mathrm{d}U_\mathrm{D}} \left(\frac{\tau}{U_\mathrm{T}} \cdot I_0 \left(\mathrm{e}^{\frac{U_\mathrm{D}}{U_\mathrm{T}}} - 1 \right) \right) \cdot \frac{\mathrm{d}U_\mathrm{D}}{\mathrm{d}t} \\ &= \left(\frac{\tau}{U_\mathrm{T}^2} \cdot I_0 (\mathrm{e}^{\frac{U_\mathrm{D}}{U_\mathrm{T}}}) \right) \cdot \frac{\mathrm{d}U_\mathrm{D}}{\mathrm{d}t}. \end{aligned} \tag{35}$$

The total capacitor current results from the parallel grouping of junction and diffusion capacitances to $C_\mathrm{P} = C_\mathrm{j} + C_\mathrm{d}$ and therefore is the sum of I_Cj and I_Cd:

$$I_\mathrm{C} = I_\mathrm{Cj} + I_\mathrm{Cd} \tag{36}$$

4.2.4 Inputs/Outputs/State Variables in EMT Simulation

Table 4.1: Input parameters on EMT Page (*ElmPvpanel*)

Parameter	Unit	Default	Description	Range	Symbol
iopt_useC		0	Consider module capacitance		
iopt_Cpmod		0	Model of Cp		
Cp_const	µF	50	Parallel capacitance	$C_\mathrm{P} \geq 0$	C_P
Cj0	µF	11.7	Zero-voltage junction capacitance	$C_\mathrm{j0} \geq 0$	C_j0
Phi0	V	0.82	Zero-voltage junction potential	$\phi_0 \geq 0$	ϕ_0
tau	µs	5.76	Mean carrier lifetime	$\tau \geq 0$	τ

Table 4.2: Input Signals in EMT Simulation

Input Signal	Unit	Description	Symbol
Egpv	W/m^2	Global irradiation on PV panel	$E_\mathrm{g,pv}$

Table 4.3: State variables in EMT Simulation

State	Unit	Type	Description	Symbol
BP_BLOCKED		discrete	Switching state of bypass diode	
UD	V	continuous	Diode voltage	U_D

5 References

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